Session 5 · Deck 5.A2

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The Growth Model to First Order

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Where We Are

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Block A — perturbation: 5.A1 the idea · **5.A2** the growth model to first order · 5.A3 which solution, higher order, distributions.

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Roadmap: Four Parts

5.A1's six steps, carried out on the one model whose answer we know. By the end the first-order solution is four numbers from four equations.

Part	Topic	The question it answers
1	The model, and what we know exactly	Which model, and what benchmark do we grade against?
2	The risk parameter and the expansion	How does risk become something we can differentiate?
3	Differentiate the identity	Where do the equations for the coefficients come from?
4	Certainty equivalence; the open question	Why are two unknowns exactly zero, and what is left open?

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Idea

Steps 1–5 of 5.A1's table, in order: write $F = 0$, pick the point, write the series, differentiate, solve. Step 6 is the next deck.

The stochastic growth model with log utility and full depreciation: states (k, z), controls (c, k'), and two equilibrium conditions.

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln c_t \quad \text{s.t.} \quad c_t + k_{t+1} = e^{z_t} k_t^{\alpha},$$

$$z_t = \rho z_{t-1} + \sigma \varepsilon_t, \quad \varepsilon_t \sim N(0,1)$$

Equilibrium conditions:

$$\underbrace{\frac{1}{c_t}}_{\text{marginal utility today}} = \beta\, \mathbb{E}_t\Big[\underbrace{\frac{1}{c_{t+1}}}_{\text{tomorrow}} \underbrace{\alpha e^{z_{t+1}} k_{t+1}^{\alpha-1}}_{\text{return on capital}} \Big]$$

$$c_t + k_{t+1} = e^{z_t} k_t^{\alpha}$$

States (k_t, z_t), inherited; **controls** (c_t, k_{t+1}), chosen. We want the **policy functions** $c(k, z)$, $k'(k, z)$.

Idea

Full depreciation ($\delta = 1$) and log utility: the Brock–Mirman benchmark of 1.B1, the one model in this course whose policy has a closed form. Calibration $\alpha = 0.36$, $\beta = 0.95$ (Session 3).

Example

$\delta = 1$ is diagnostic, not realistic. 5.A3 moves to the quarterly $\delta = 0.025$, where no closed form exists.

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Guess and verify gives the exact policy and the steady state. The steady state is the point we expand around; the policy is the answer we grade against.

$$c_t = (1-\alpha\beta)\, e^{z_t} k_t^{\alpha}$$
$$k_{t+1} = \alpha\beta\, e^{z_t} k_t^{\alpha}$$

A constant fraction $\alpha\beta$ of output is saved. Deterministic steady state ($z = 0$, $k_{t+1} = k_t$):

$$k^* = (\alpha\beta)^{\frac{1}{1-\alpha}} = 0.1870, \qquad c^* = k^{*\alpha} - k^*, \qquad z^* = 0.$$

Perturbation needs only the **steady state**, which almost every model has, not the closed-form **policy**, which almost none has.

Example: a check, not a test

In logs the exact policy is $\ln k_{t+1} = \ln(\alpha\beta) + z_t + \alpha \ln k_t$: *exactly linear*. A first-order expansion in logs must return the slopes α and 1 exactly, not approximately. Solved numerically, it does, to 10^{-16}.

Pitfall

That certifies the algebra and the code, not the accuracy: the error is zero only because this model is log-linear.

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Ledger brock_mirman: **slopes** 0.36 **and** 1.00; **deviation from the exact policy** 1.1×10^{-16}.

The Goal, Written as Functional Equations

Put the unknown policies into the equilibrium conditions. The unknowns are now functions, and one of them appears evaluated at next period's state.

$$\frac{1}{c(k_t, z_t)} = \beta\, \mathbb{E}_t \left[\frac{\alpha\, e^{\rho z_t + \sigma \varepsilon_{t+1}}\, k(k_t, z_t)^{\alpha - 1}}{c\big(\underbrace{k(k_t, z_t)}_{\text{tomorrow's capital}},\ \underbrace{\rho z_t + \sigma \varepsilon_{t+1}}_{\text{tomorrow's shock}}\big)} \right]$$

$$c(k_t, z_t) + k(k_t, z_t) = e^{z_t} k_t^{\alpha}$$

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Idea

c appears *evaluated at next period's state*, which is itself the unknown $k(\cdot)$. That nesting is why this is a functional equation and not an ordinary one.

Once $c(\cdot)$ and $k(\cdot)$ are known, everything follows: simulate, compute moments, produce impulse responses, do welfare.

A Parameter That Scales the Risk

Uncertainty is not a state variable, so we cannot expand in it. The device: a dial χ on the shock, and expand around the dial set to zero.

Multiply the shock by a parameter χ:

$$z_t = \rho z_{t-1} + \underbrace{\chi}_{\text{dial}} \sigma \varepsilon_t$$

- $\chi = 1$: the model we care about.
- $\chi = 0$: the **deterministic** economy. No uncertainty, a steady state we can compute, policies we can find directly.

The policies become $c(k_t, z_t; \chi)$ and $k(k_t, z_t; \chi)$, and χ is a third argument to differentiate with respect to.

Why around $\chi = 0$? It is the one point where the answer is available. The expansion then measures how a *small amount of risk* changes behaviour — exactly the economic question.

Idea

Risk has been turned into a *parameter we can differentiate*. Everything in 5.A3's second-order block is a consequence of this one move.

Notation

The literature writes λ for this dial; this course reserves λ for the invariant distribution (2.A3), so we write χ.

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The Expansion, Written Out

Build the series at $(k^*, 0; 0)$ and use it at $\chi = 1$. Every coefficient is a derivative of the true policy at the steady state.

With $\hat{k}_t = k_t - k^*$, and every derivative evaluated at $(k^*, 0; 0)$:

$$c_t = c(k_t, z_t; 1) = \underbrace{c^*}_{\text{level}} + \underbrace{c_k \hat{k}_t + c_z z_t + c_\chi}_{\text{first order: slopes}} + \tfrac{1}{2}\Big(\underbrace{c_{kk}\hat{k}_t^2 + 2c_{kz}\hat{k}_t z_t + c_{zz} z_t^2}_{\text{curvature in the states}} + \underbrace{2c_{k\chi}\hat{k}_t + 2c_{z\chi} z_t + c_{\chi\chi}}_{\text{how risk changes the rule}} \Big) + \cdots$$

and the same for $k_{t+1} = k(k_t, z_t; 1)$ with $k_k, k_z, k_\chi, \ldots$ in place of $c_k, c_z, c_\chi, \ldots$

Idea

First order keeps the slopes: c_k, c_z, c_χ and k_k, k_z, k_χ — **six unknowns**. Part 3 finds their equations; Part 4 shows two of them are zero.

Example: built at 0, used at 1

Where χ stood there is now 1, so c_χ and $\frac{1}{2}c_{\chi\chi}$ are **constants that shift the policy**. $c_{\chi\chi}$ is the one that will carry precautionary saving (5.A3).

Mixed partials of a smooth function are equal ($c_{kz} = c_{zk}$), which is why each cross term appears once with a factor 2.

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3 · Package the Conditions as $F(k_t, z_t; \chi) = 0$

One object with two components, so that "differentiate and match" is mechanical rather than a case-by-case argument.

$$F(k_t, z_t; \chi) = \mathbb{E}_t \left(\begin{array}{c} \dfrac{1}{c(k_t, z_t; \chi)} - \beta \dfrac{\alpha\, e^{\rho z_t + \chi\sigma\varepsilon_{t+1}} \big(k(k_t, z_t; \chi)\big)^{\alpha-1}}{c\Big(k(k_t, z_t; \chi),\ \rho z_t + \chi\sigma\varepsilon_{t+1}; \chi\Big)} \\ c(k_t, z_t; \chi) + k(k_t, z_t; \chi) - e^{z_t} k_t^{\alpha} \end{array} \right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Inside the expectation sits **one function of six arguments**, $H(c_t,\ c_{t+1},\ k_t,\ k_{t+1},\ z_t;\ \chi)$, with $c_t = c(k_t, z_t; \chi)$, $k_{t+1} = k(k_t, z_t; \chi)$ and $c_{t+1} = c\big(k_{t+1},\ \rho z_t + \chi\sigma\varepsilon_{t+1}; \chi\big)$.

Idea

First component: the Euler equation. Second: the resource constraint. Both must be zero, so F is a 2-vector and every equation we derive from it comes in pairs.

Notation

H_i is the response of H (both components) to its i-th argument, at the steady state: H_1 for c_t, H_2 for c_{t+1}, H_3 for k_t, H_4 for k_{t+1}, H_5 for z_t, H_6 for χ. **Known numbers.**

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3 · An Identity, So Its Derivatives Vanish

With the true policies substituted in, $F = 0$ holds for every (k, z, χ) near the steady state. A function that is zero everywhere has zero derivatives, and those are the equations.

$$F \equiv 0 \implies F_k = F_z = F_\chi = 0, \quad F_{kk} = F_{kz} = \cdots = 0,$$

all evaluated at $(k^*, 0; 0)$. The unknown coefficients $c_k, c_z, \ldots$ sit *inside* these derivatives, through the chain rule. Setting each derivative to zero therefore gives **equations for them**.

First order needs (F_k, F_z, F_χ). Second order also needs $(F_{kk}, F_{kz}, \ldots)$, and so on. This is 5.A1's $G \equiv 0$, with F in place of G.

Idea: why not $F = 0$ itself?

$F = 0$ at the expansion point is the **zeroth-order** condition: there $\hat{k}_t = z_t = \chi = 0$, every slope term multiplies zero, and what survives is the steady state we already knew. The slopes come from the derivatives.

Contrast: Block B

Projection does the opposite: it imposes $F = 0$ at **many points**, so the coefficients appear directly and no derivative is taken.

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Differentiate $F = 0$ in k_t, z_t and χ: each term is a known response H_i times how argument i moves.

$$F_k = \underbrace{H_1 c_k}_{c_t \text{ moves}} + \underbrace{H_2\, c_k k_k}_{c_{t+1} \text{ moves via } k_{t+1}} + \underbrace{H_3}_{k_t \text{ itself}} + \underbrace{H_4 k_k}_{k_{t+1} \text{ moves}} = 0$$

$$F_z = H_1 c_z + H_2\big(c_k k_z + c_z \rho\big) + H_4 k_z + H_5 = 0$$

$$F_\chi = H_1 c_\chi + H_2\big(c_k k_\chi + c_\chi\big) + H_4 k_\chi + \mathbb{E}_t\big[\ \underbrace{H_2\, c_z\, \sigma \varepsilon_{t+1} + H_6}_{\text{the part with no unknown}}\ \big] = 0$$

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Idea

c_{t+1} moves two ways: through tomorrow's capital $k_{t+1} = k(k_t, z_t; \chi)$ and through tomorrow's shock $z_{t+1} = \rho z_t + \chi \sigma \varepsilon_{t+1}$ — so H_2 multiplies a sum ($c_k k_z$ and $c_z \rho$ in F_z). Each line is a 2-vector.

Pitfall

F_k contains $H_2 c_k k_k$, a **product of two unknowns**: the system for the first-order coefficients is quadratic, so it can have more than one solution. That is 5.A3's question.

Count. The state derivatives give four scalar equations for four unknowns; the χ-derivative handles the other two on its own.

$F_k = 0$ and $F_z = 0$, two components each: **four equations** for the four unknowns c_k, c_z, k_k, k_z. Square.

The remaining two unknowns, c_χ and k_χ, appear only in $F_\chi = 0$:

$$(H_1 + H_2)\, c_\chi + (H_2 c_k + H_4)\, k_\chi + \mathbb{E}_t[\cdots] = 0,$$

two equations, **linear** in (c_χ, k_χ) once c_k is known. If the bracket is zero the system is **homogeneous**, and its only solution is

$$c_\chi = k_\chi = 0.$$

Idea

"Homogeneous" means every term carries an unknown. A linear system with no constant term is solved by setting all unknowns to zero, and (generically) by nothing else.

So the first-order solution is **four numbers**, and the shock scale χ appears in none of them. Why the bracket is zero: next frame.

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Why the χ-Equation Has No Constant

χ reaches the equilibrium conditions only through tomorrow's shock. Every such term is proportional to ε_{t+1}, whose expected value is zero.

χ enters only through $z_{t+1} = \rho z_t + \chi\sigma\varepsilon_{t+1}$, in two places:

- **directly**, in $e^{z_{t+1}}$ inside the Euler equation. $H_6 = \partial H/\partial\chi$ has one entry per row of H:

$$H_6 = \begin{pmatrix} -\sigma\varepsilon_{t+1}/c^* \\ 0 \end{pmatrix} \begin{matrix} \leftarrow \text{Euler row} \\ \leftarrow \text{resource row} \end{matrix}$$

  Euler: $\partial e^{\chi\sigma\varepsilon_{t+1}}/\partial\chi$ brings down $\sigma\varepsilon_{t+1}$, and $\beta\alpha k^{*\alpha-1} = 1$. Resource: no χ in it. Unlike $H_1, \ldots, H_5$, H_6 is **random**.

- **through tomorrow's policy**, $c_{t+1} = c(k_{t+1}, z_{t+1}; \chi)$, which contributes $H_2 c_z \sigma\varepsilon_{t+1}$.

Both are a fixed vector times ε_{t+1}, so

$$\mathbb{E}_t\left[H_2 c_z \sigma\varepsilon_{t+1} + H_6\right] = \left[H_2 c_z - \begin{pmatrix} 1/c^* \\ 0 \end{pmatrix}\right] \sigma \underbrace{\mathbb{E}_t[\varepsilon_{t+1}]}_{=0} = 0.$$

Idea

A small *linear* change in the shock size moves the economy up as often as down, and the expectation averages it away. Risk enters through the **variance**, quadratic in χ: it first appears in $c_{\chi\chi}$ (5.A3).

Pitfall

$\mathbb{E}_t[\varepsilon_{t+1}] = 0$ does all the work. Give the innovation a non-zero mean and the bracket is no longer zero.

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Certainty Equivalence: What It Means

$c_\chi = k_\chi = 0$: the shock scale enters no first-order coefficient. To a linear approximation, an agent facing risk behaves exactly like one who faces none.

The first-order solution is a pair of linear rules,

$$\hat{c}_t = a_k \hat{k}_t + a_z z_t, \qquad \hat{k}_{t+1} = b_k \hat{k}_t + b_z z_t,$$

with a_k, a_z, b_k, b_z the four numbers of Part 3. Set σ ten times larger and the four numbers do not change.

What the linear solution therefore cannot see:

- precautionary saving — the extra saving a risky future induces;
- risk premia, and the welfare cost of uncertainty.

All three are *identically zero* at first order — not small, zero.

Idea

Not because the model lacks precaution, but because a straight line cannot bend. Precaution lives in the curvature of marginal utility (2.A1's prudence), and curvature is a second-order object.

First order describes **dynamics** well and **risk** not at all. Whatever a linearized model says about uncertainty was assumed, not computed.

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The four equations are quadratic in the unknowns, so they can have more than one solution. Counting equations says the system is square; it does not say which solution is the economy's.

Substitute the linear rules into the equilibrium conditions and what remains is a **linear rational-expectations system**: today's variables, tomorrow's expected variables, and the shock. Such a system can have

- no stable solution,
- many stable solutions (indeterminacy),
- or exactly one.

Nothing in Part 3's counting distinguishes these.

Example

1.B1 drew it: from any k_0 the saddle path picks exactly one c_0; every other c_0 runs off to zero consumption or zero capital. The same choice is hiding in the quadratic.

Blanchard–Kahn (5.A3) tells you which root to keep, and whether keeping exactly one is possible at all.

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Takeaways

- **Part 1.** The Brock–Mirman model: log utility, $\delta = 1$, closed-form policy, $k^* = 0.1870$. Perturbation needs only the steady state, which is why it generalizes; the closed form is for checking.
- **Part 2.** The dial χ on the shock, $z' = \rho z + \chi \sigma \varepsilon'$, makes risk a parameter we can differentiate. Expand at $\chi = 0$, use at $\chi = 1$; six first-order unknowns.
- **Part 3.** With the true policies in, $F \equiv 0$, so $F_k = F_z = F_\chi = 0$. Each term is a known H_i times how an argument moves. Four scalar equations for c_k, c_z, k_k, k_z.
- **Part 4.** $F_\chi = 0$ is homogeneous because every χ-term is proportional to ε_{t+1} and $\mathbb{E}_t[\varepsilon_{t+1}] = 0$: **certainty equivalence**, $c_\chi = k_\chi = 0$. The system is quadratic, so which solution is still undecided.

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Bridge: Which Root, and Is There One?

We have a square system with a quadratic term, hence more than one solution, and only one of them is economically meaningful. Picking it is not a numerical detail — it is the difference between a model with a unique equilibrium and one with a continuum.

After that, two extensions of the same recipe: one more order of derivatives, which is where risk finally enters; and one more kind of expansion point, a whole distribution.

Next (5.A3)

Blanchard–Kahn: eigenvalues, predetermined versus jump variables, and the count that decides uniqueness. How the computer finds the stable solution. Then second order and the risk correction, and perturbing around a distribution.

Keep the closed form in view: everything 5.A3 computes on this model can be checked against α and 1.

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